The Influence of Income on Cooking Fuel Expenditure

This report has been submitted as part of the end term assessment for Cross-Section & Panel Data Econometrics

Submitted By: Yash Kumar Singh

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Dr. B.R. Ambedkar University, Delhi

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Introduction

Energy is a fundamental pillar of human development and it is also a critical target under the United Nations sustainable development goals. Over the few years India has made remarkable transformations in the energy supply infrastructure by providing 97.4% of households with electricity but the transition to clean cooking fuel still remains an unfinished challenge. As of late 2024, despite the expansion of LPG networks upper portion of Indian population still continues to rely on solid fuels which reveals a major energy transition gap that affects the rural and lower income households unfairly. The Pradhan Mantri Ujjwala Yojna which was launched in 2016 claims to aggressively address this gap. The scheme provided 10.33 crore LPG connections to economically disadvantaged households. But the access to these LPG cylinders does not guarantee the adoption within rural and low-income households while the connections have increased but there exists a refill gap due to affordability constraints. Wood, dung and crop residue are commonly used solid fuel and its combustion has severe implications for both public health and social equity. Economic and social theory suggests that as households' income rises their consumption shifts from inferior goods to normal or luxury goods. Not only income but other socio-economic factors also like caste, education, and gender differences with in the household. (Salman Haider, 2022) examined the socio-economic determinants of cooking fuel expenditure in the context of Uttar Pradesh (UP). It did not make causal claims but it provided a descriptive and regression analysis. For this purpose, nationally representative survey data from consumer pyramids household survey from 2014 May to December 2019 has been used. This specific Study is built upon this work only but it extends the analysis not only to Uttar Pradesh but all of the states of India which may reveal different energy expenditure patterns.

Data & Methodology

Data Collection

This study is based on the survey data provided by the Centre for Monitoring Indian Economy called as Consumer Pyramids Household Survey (CPHS) database. It is an extensive panel household survey covering over 170,000 Indian households across 27 states and 514 districts. Each household is interviewed thrice every year (three waves each year). Income and consumption data for four preceding months are collected during each wave, so CPHS have monthly data on income and consumption. This study used data from May 2014 (second wave) as there is over-sampling in wave 1 (Jan to April 2014) and then adjusted in the second wave. Hence the sample is selected from May 2014 to Dec-2019 (Salman Haider, 2022).

The CPHS data comes with two necessary information about consumption & income and household information. The monthly consumption and income information is stored under MCI dataset whereas the household information is stored under HI dataset and both of them are in parquet form to be handled efficiently.

Methodology

Both of the data sets mentioned above that is MCI and HI were merged using household Id and year. The resulting data set is used as a panel which includes all the household across the states in India. This allows a national level analysis of cooking fuel expenditure patterns. The variables present in the panel are discussed in the next section. The empirical analysis was conducted in two parts. At first, we performed a exploratory data analysis which include state & region wise expenditure on cooking fuel, year wise expenditure on cooking & electricity expenditure and lastly a correlation matrix to provide linear associations between the variables included in the analysis. In the next part we quantify the associations using different regression techniques like Fixed Effects, Random Effects, Pooled regression, Fixed Effect Least Square Dummy Variable Regression (FELSDV) etc. Then various diagnostic tests like Breusch–Pagan Test (Heteroskedasticity), Durbin Watson Test (Autocorrelation).

Variables

Table 1 represents all of the variables used along with their description.

Table 1: Variable Description

Variable	Description
cook_pc	This is the per capita cooking expenditure of a household
totexp_pc	Per capita total expenditure of a household
ln_cook	Log of the expenditure on cooking fuel and this is also the dependent variable for this study
ln_totexp	Log of the total expenditure and the explanatory variable of this study used as a proxy for household income supported by literature.
M_EXP_ELECTRICITY	Monthly household expenditure on electricity
M_EXP_COOKING_FUEL	Monthly household expenditure on cooking fuel
SIZE_NUM	Number of people in a household taken as a midpoint for responses provided as a range, control variable

REGION_TYPE_x	Control variable for urban and rural areas
STATE	Control variable for the states

Model Specification

The baseline model used is a two-way fixed effects model in the following form:

$$ln_cook = \alpha i + \beta ln(totexp) + \gamma Xit + \epsilon it,$$

Where X is a vector of the control variables, i is the household index, α is the intercept and β is the coefficient of the explanatory variables. The choice Fixed Effects model is justified to take different household characteristics differ significantly according to the time invariant factors into consideration and gives consistent estimates. Also, a two - way fixed effects model is used because ignoring time effects can lead to biased coefficients.

Empirical Results

Descriptive Statistics & Correlation Analysis

Summary statistics of the key variables is showed in table 2. Table 2 gives the summary statistics for the main variables in this analysis. Household cooking expenditure per capita, which is cook_pc, shows a lot of variation. The mean comes in at 112.56, and the standard deviation points to a pretty wide spread. Total household expenditure per capita, totexp_pc, and its log version, ln_totexp, also display quite a bit of heterogeneity. That reflects the differences in how households stand economically. The log-transformed variables, like ln_cook and ln_totexp, have distributions that are more squeezed together. This suggests less skewness compared to the original raw data. Household spending on electricity, M_EXP_ELECTRICITY, has high variability overall. The median stays small, but the maximum value reaches pretty high. Average household size, SIZE_NUM, works out to about four members. Variability there remains relatively low. In general, these descriptive stats highlight the notable dispersion in how households handle consumption and spending patterns.

Table 2: Summary Statistics

Variable	Count	Mean	Standard Deviation	25%	50%	75%	Max
cook_pc	15272380	112.56	140.27	0	9.85	187.5	4833

totexp_pc	15272380	2776.2	10308.68	1326.8	2379.4	3732.5	19506
ln_cook	12621550	3.75	2.308	0	4.84	5.36	8.48
ln_totexp	12621550	7.94	5.82	7.54	7.92	8.32	16.78
M_EXP_ELECTRICITY	19865740	278	474.97	0	2	4.2	4500
SIZE_NUM	15272380	3.94	1.59	3	4	5	17.5

Table 3 shows the average monthly fuel expenditure. The variation is less but not constant over the month. The expenditure starts increasing after May maybe due to seasonal change.

Table 3: Mean Cooking Fuel Expenditure by Month

Month	Mean Expenditure (₹)
January	108.60
February	109.75
March	113.70
April	108.98
May	107.52
June	113.41
July	115.81
August	119.48
September	119.89
October	116.14
November	113.91
December	104.89

Table 4 representing mean of total expenditure per capita does not show much variation and the increase can be observed after the month of May similar to Table 3.

Table 4: Monthly Mean of total expenditure per capita

Month	Mean Total Expenditure (₹)
January	2705.26
February	2722.37
March	2753.76
April	2753.59
May	2598.89
June	2799.28
July	2761.21

August	2887.48
September	2878.05
October	2971.55
November	2904.36
December	2598.47

Table 5 aims to examine regional differences between rural and urban areas across their expenditure on electricity and cooking fuel. The difference demonstrates the financial gap between the regions. The cost of living is slightly high in the urban areas as compared to those living in rural areas.

Table 5: Rural vs Urban Expenditure Differences

Region	Household Size	Electricity Expenditure (₹)	Cooking Fuel Expenditure (₹)	Total Expenditure (₹)
Rural	4.12	181.41	100.16	2260.98
Urban	3.87	327.87	118.37	3017.21

Table 6 represent the average per household energy expenditure and it is visually represented in figure 1. The disparities are clearly visible between the regions.

Table 6: Per Household Expenditure (Rural vs Urban)

Region	Cooking Fuel Expenditure (₹ per household)	Electricity Expenditure (₹ per household)
Rural	313.42	181.41
Urban	375.10	327.87

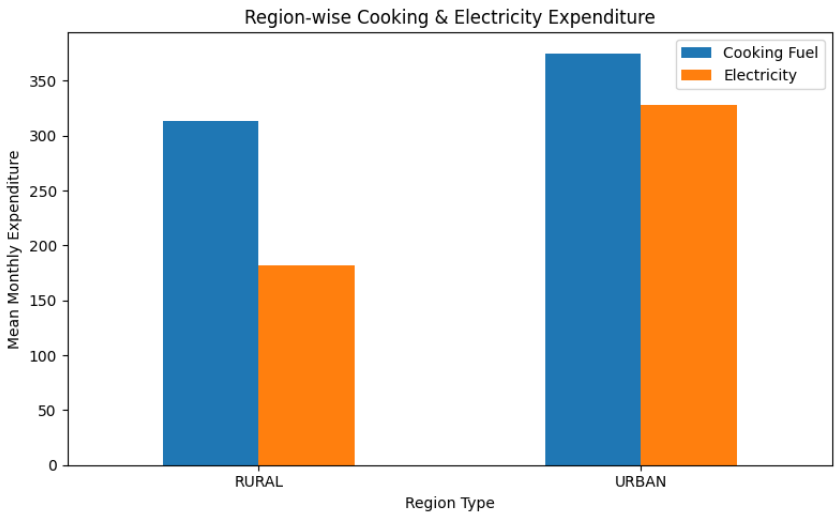

Figure 1: Mean Monthly Energy Expenditure (Rural vs Urban)

State wise analysis was also performed to identify causal relationships across all states of India. Table 7 highlights major variation in household size, electricity spending, cooking fuel expenditure, and total per capita expenditure across Indian states. Smaller and urban states such as Chandigarh, Delhi and Punjab show significantly higher electricity and total expenditure spending. On the other hand, states like Bihar, Odisha, and Jharkhand show lower energy spending. Overall, the table shows a clear inter-state disparity in energy and household economic status.

Table 7: State wise Descriptive Statistics

State	HH Size	Electricity Expenditure (₹)	Cooking Fuel Expenditure per capita (₹)	Total Expenditure per capita (₹)
Andhra Pradesh	3.06	209.12	94.77	2975.47
Assam	3.82	173.70	140.07	2286.23
Bihar	4.44	171.90	122.01	1802.00
Chandigarh	3.46	915.98	164.83	5654.85
Chhattisgarh	4.10	230.64	117.85	2303.51
Delhi	4.97	646.23	139.42	3846.00
Goa	3.58	367.26	116.96	4793.72
Gujarat	4.34	210.49	54.55	2345.35
Haryana	3.81	357.10	175.55	4257.22
Himachal Pradesh	3.74	235.02	142.45	3624.26
Jammu & Kashmir	4.33	265.54	135.72	3280.31
Jharkhand	4.10	166.84	146.89	2238.02
Karnataka	3.63	161.15	118.76	2644.75
Kerala	3.59	189.46	83.63	3574.61
Madhya Pradesh	4.05	277.53	86.81	2316.06
Maharashtra	3.97	434.33	126.20	3026.92
Meghalaya	5.34	265.65	162.94	3868.69

Odisha	3.72	147.86	97.57	2092.05
Puducherry	3.54	197.94	79.27	3234.27
Punjab	3.50	329.47	168.42	4277.78
Rajasthan	4.05	389.40	85.21	2893.71
Sikkim	4.20	12.81	102.56	2866.84
Tamil Nadu	3.38	83.10	88.17	2774.34
Telangana	3.15	220.52	78.93	3508.37
Tripura	3.41	237.57	219.29	3112.48
Uttar Pradesh	4.54	390.57	118.35	2157.78
Uttarakhand	3.47	365.29	154.77	4995.34
West Bengal	4.02	231.52	108.87	2129.47

Figure 2 demonstrates the discussed disparities above clearly.

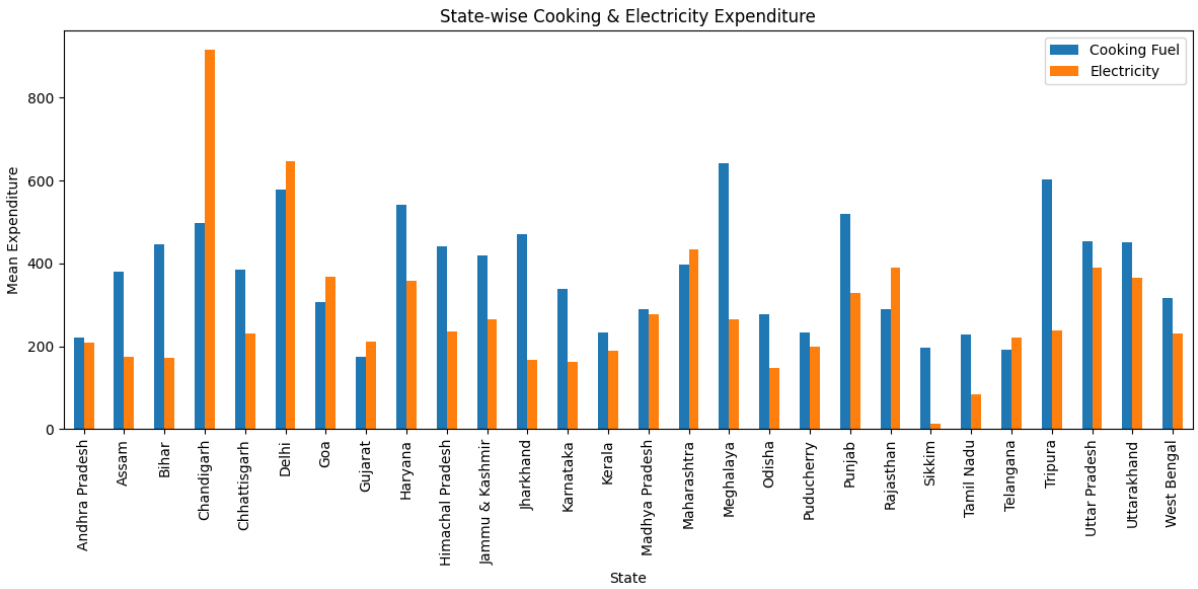

Figure 2: State wise average expenditure on Energy

Figure 3 represents a year wise analysis of cooking fuel and electricity expenditure from the year 2014 to 2024. A substantial growth can be observed from year 2014 to year 2018. The energy spending can be seen declined in the year 2020 perhaps due to COVID-19 pandemic. This also supports the time fixed effects model for consistent estimates.

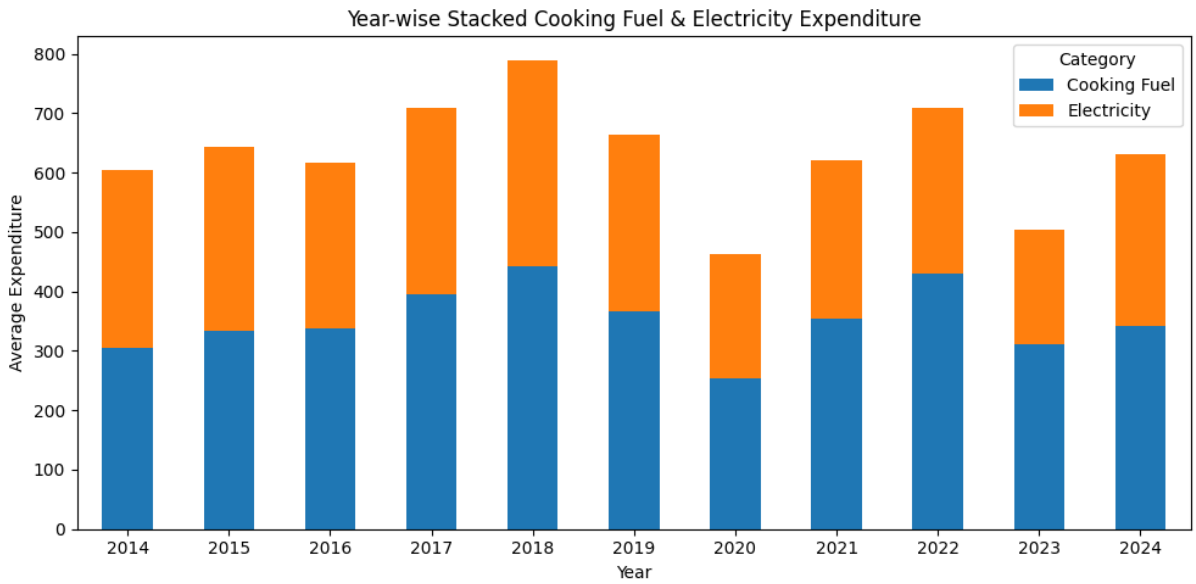

Figure 3: Year wise Energy Expenditure (2014-2024)

To analyse the linear association between the key variables used a correlation analysis was performed. The results are demonstrated through the correlation heatmap in figure 4.

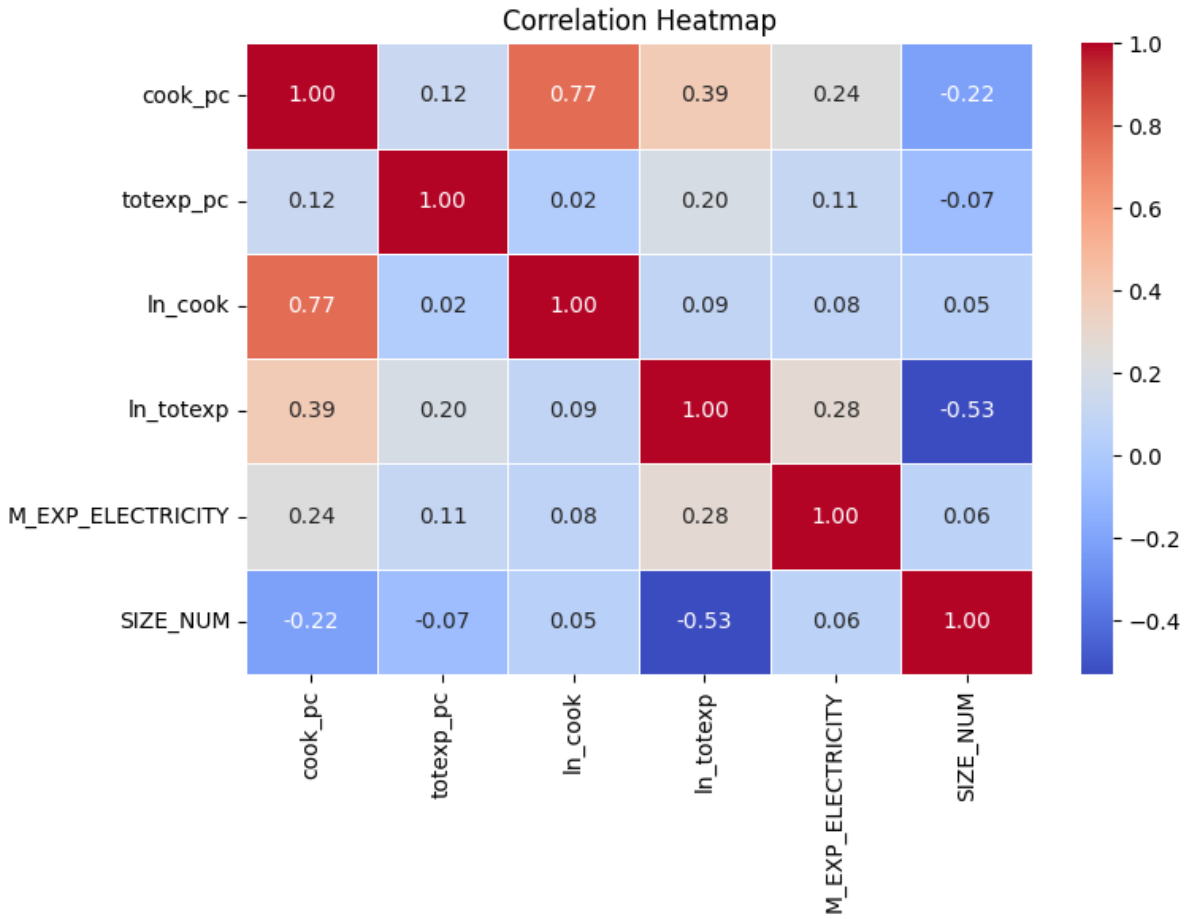

Figure 4: Correlation Heatmap

The main focus of this paper i.e. the association between per capita cooking expenditure and per capita total expenditure is positive but weak ($r = 0.124$) indicating households with higher consumption tends to spend more on cooking fuel than those with lower level of consumption. On taking log transformation of total expenditure the association has strengthened ($r = 0.387$). Other correlations are generally weak, indicating limited linear associations among the rest of the variables.

The descriptive statistics just highlights patterns in the data, but to explain the determinants of energy expenditure and to understand such relationship more rigorously, this analysis proceed to econometric analysis which allows us to quantify the influence of household features while controlling for other factors.

Empirical Analysis

Table 8 presents the regression results for the determinants of household cooking fuel expenditure(ln_cook). The analysis started from a baseline Pooled OLS to more advanced Fixed Effects models to control for unobserved heterogeneity.

Table 8: Regression Results: Determinants of Cooking Fuel Expenditure (ln_cook)

Variables	**(1) Pooled OLS**	**(2) Entity FE**	**(3) Time FE**	**(4) Two-Way FE**	**(5) Random Effects**
Log Total Exp.	0.630***	0.805***	0.886***	1.321***	0.449***
	(0.001)	(0.004)	(0.005)	(0.0048)	(0.000)
Household Size	0.188***	0.172***	0.206***	0.189***	0.083***
	(0.000)	(0.002)	(0.002)	(0.0018)	(0.001)
Electricity Exp.	—	-4.286-e05***	7.714e-05***	-9.313e-05***	4.04e-05***
		(3.672e-06)	(4.348e-06)	(3.634e-06)	(1.39e-06)
Constant	-1.999***	—	—	—	—
Observations	12,621,553	12,621,553	12,621,553	12,621,553	12,621,553
R^2 (Overall/Within)	0.020	0.021	0.034	0.043	0.148
Effects	None	Entity	Time	Entity & Time	Random
SE Type	Non-Robust	Clustered	Clustered	Clustered	Unadjusted

Dependent Variable: Log Cooking Fuel Expenditure (ln_cook).

Standard errors in parentheses. ***$p < 0.01$.

Models (2), (3) and (4) use standard errors clustered at the entity level to control for serial correlation.

Source: Authors' calculations based on CPHS.

The results from the Pooled OLS (1) model indicate a positive and significant relationship between total household expenditure and cooking fuel expenditure. The resulting coefficient suggests that a 1% increase in total household expenditure makes a 0.63% increase in cooking fuel expenditure on average holding all other factors constant. Additionally, household size (*SIZE_NUM*) is positively correlated with fuel expenditure, with each additional member increasing expenditure by approximately 18.8%.

But a pooled OLS can lead to OVB problem because it relies on the assumption that no unobserved household and time features are there. But usually there exist such different characteristics which often are unobserved hence making the estimates likely biased.

To address the potential OVB, we ran an Entity Fixed Effects regression (2). It controls time-invariant household features, making the estimate change to 0.80. This suggests that the Pooled OLS estimates were biased. But it is as important to control time specific shock also for e.g., inflation, COVID 19 etc.

And to address such time shocks a Time Fixed Effects regression (3) was performed and the estimates increased slightly to 0.88. This suggests that time shocks were influencing the observed relationship in the baseline model.

Next a Two-way Fixed Effect regression (4) was performed which controls both entity specific heterogeneity and time shocks. In this model, the estimate of total expenditure increased to 1.32, suggesting that a 1% increase in total expenditure leads to a 1.32% increase in cooking fuel expenditure. All of the estimates were statistically significant at 5% level of significance. The F-test for poolability rejects the use of pooled OLS and confirms that these household specific effects are statistically relevant and that pooled OLS will produce biased estimates.

A Random Effect (5) model assumes that household features are uncorrelated with the explanatory variables. The results suggests that a 1% increase in a household's total annual expenditure is associated with a 0.45% increase in cooking fuel expenditure. Also, an increase in household size by 1 member leads to 8.3% increase in cooking fuel spending. Though the explanatory power (R^2) is high but RE depends on the assumption that unobserved features are not correlated with the independent variables.

Model Comparison Tests

The following snapshot compares Pooled OLS with the Entity Fixed Effect model and it supports the fact that Entity FE is a better choice.

```
                    Model Comparison
=====================================================
                                Pooled      FE_entity
-----------------------------------------------------
Dep. Variable                  ln_cook        ln_cook
Estimator                    PooledOLS       PanelOLS
No. Observations              12621553       12621553
Cov. Est.                   Unadjusted     Unadjusted
R-squared                       0.0210         0.0210
R-Squared (Within)              0.0181         0.0210
R-Squared (Between)             0.0088         0.2477
R-Squared (Overall)             0.0210         0.1676
F-statistic                  9.042e+04      8.852e+04
P-value (F-stat)                0.0000         0.0000
====================     ============   ============
const                          -1.6022
                             (-131.76)
ln_totexp                       0.5794         0.8048
                              (413.53)       (503.84)
SIZE_NUM                        0.1755         0.1716
                              (356.57)       (250.98)
M_EXP_ELECTRICITY               0.0001     -4.286e-05
                              (97.912)      (-29.737)
====================== ============== ==============
Effects                                        Entity
```

Comparison of Entity only and Time only FE model.

```
                    Model Comparison
=====================================================
                            FE_no_time   FE_with_time
-----------------------------------------------------
Dep. Variable                  ln_cook        ln_cook
Estimator                     PanelOLS       PanelOLS
No. Observations              12621553       12621553
Cov. Est.                   Unadjusted     Unadjusted
R-squared                       0.0210         0.0430
R-Squared (Within)              0.0210         0.0109
R-Squared (Between)             0.2477        -2.4892
R-Squared (Overall)             0.1676        -2.1428
F-statistic                  8.852e+04      1.857e+05
P-value (F-stat)                0.0000         0.0000
====================     ============   ============
ln_totexp                       0.8048         1.3214
                              (503.84)       (733.85)
SIZE_NUM                        0.1716         0.1893
                              (250.98)       (280.13)
M_EXP_ELECTRICITY           -4.286e-05     -9.313e-05
                             (-29.737)      (-65.575)
====================== ============== ==============
Effects                         Entity         Entity
                                                 Time
```

Following is the comparison between the FE and RE regression model.

```
                    Model Comparison
=========================================================
                                  FE                RE
---------------------------------------------------------
Dep. Variable                 ln_cook           ln_cook
Estimator                    PanelOLS     RandomEffects
No. Observations             12621553          12621553
Cov. Est.                   Clustered        Unadjusted
R-squared                      0.0210            0.1475
R-Squared (Within)             0.0210            0.0170
R-Squared (Between)            0.2477            0.8986
R-Squared (Overall)            0.1676            0.7287
F-statistic                 8.852e+04         7.278e+05
P-value (F-stat)               0.0000            0.0000
====================     ============   ===============
ln_totexp                      0.8048            0.4494
                             (195.69)          (1122.7)
SIZE_NUM                       0.1716            0.0825
                             (93.728)          (157.56)
M_EXP_ELECTRICITY           -4.286e-05        4.043e-05
                            (-11.671)          (29.030)
======================= ============== =================
Effects                         Entity
```

Post estimation diagnostics confirm that the Two-way FE model is the robust choice.

Data Diagnostic Tests

We conducted a Breusch-Pagan test to check for heteroskedasticity. The test yielded a significant result $LM\ stat \sim 1.08e6, p < 0.001$, leading us to reject the null hypothesis of homoskedasticity.

We also tested for serial correlation using the Durbin-Watson statistic. The resulting value of 0.96 indicates the presence of positive first-order autocorrelation.

Conclusion

The objective of this study is to investigate the determinants of the household expenditure on cooking fuel, using rich panel data from CPHS. The first part of the study focused on descriptive analysis wherein the results signalled towards socio-economic variations among the households. The urban households were spending more than the rural households, the rural households bearing a much larger burden of the expenditure. State level patterns also further identified the inequality in India in energy access and affordability. These patterns were reinforced by econometric results. Whereas the positive relationship between income and cooking fuel expenditure emerged from pooled OLS, pooling failed to take into account

unobserved household traits and time shocks. Moreover, diagnostic tests strongly rejected the pooled OLS in favour of panel methods. The random effects model provided smaller elasticities but rested on implausible assumptions about the orthogonality of unobserved household characteristics and regressors.

References

Centre for Monitoring Indian Economy. (2025). *Consumer Pyramids Household Survey (CPHS)*. CMIE. Retrieved from https://consumerpyramidsdx.cmie.com

Salman Haider (2022). Explaining household expenditure on cooking fuel: Role of income and Socio-economic status. Centre of Development Economics and Institute of Economic Growth, New Delhi, India.